

FROM POINT CLOUDS TO POLYGONAL WALLS



A Lightweight Representation of SLAM Maps for Real-Time XR

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MOTIVATION

Robots build 3D maps as they move using Simultaneous Localization and Mapping (**SLAM**) methods, but this **point clouds** are dense and unstructured for efficient real-time Extended Reality (**XR**) visualization.



OUR APPROACH

Exploit architectural priors

Indoor environments provide repetitive geometric structure:

- Planar walls
- Straight boundaries
- Shared floor & ceiling heights
- Perpendicular room intersections

Rectify individual walls

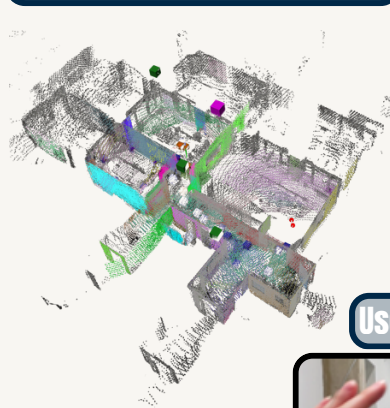
1. Project wall observation to 2D
2. Cluster/group points
3. Fill gaps & regularize perimeter

Rectify the scene

1. Align floor/ceiling
2. Snap nearby consecutive walls to consistent perpendicular intersections

SLAM Point Cloud

Incomplete, unstructured,
heavy

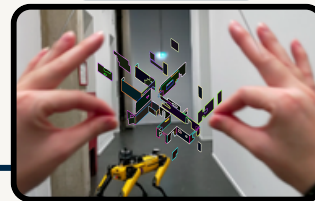


Polygonal Walls

Complete, structured,
lightweight

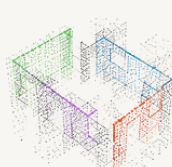


User POV in XR



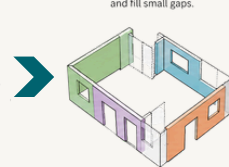
EXTRACT & ORGANIZE

Detect wall points and organize them.



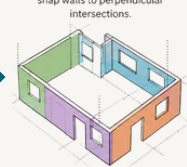
RECONSTRUCT WALLS & FILL GAPS

Build continuous walls and fill small gaps.



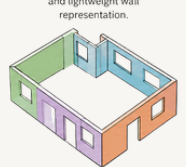
ALIGN & SNAP TO ARCHITECTURE

Align floor/ceiling levels and snap walls to perpendicular intersections.



CLEAN, STRUCTURED WALL GEOMETRY

Obtain a complete, consistent and lightweight wall representation.



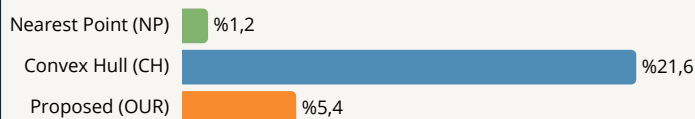
RESULTS

Coverage vs. Excess area (D1, 504 m²)

Coverage of ground-truth/real building wall area

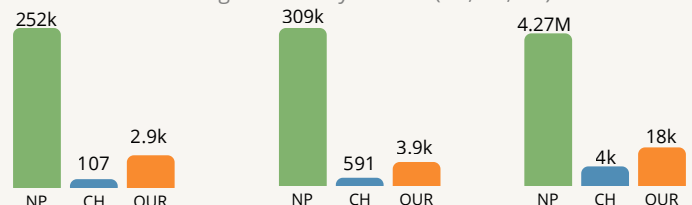


Excess (fabricated/hallucinated) area error



Mesh sizes

Triangles count by dataset (D1, D2, D3)



+50%
Increase in wall
coverage

≈133X
Fewer triangles

Enable a robot to share its live understanding of an environment as an interactive Mixed Reality map.